Transformations of y=x

1

a Yes

b No

2

a They become very large in the negative direction.

b They become very large in the positive direction.

3

a Increasing

b Equally steep

4

a No

b No

c No

d Yes

5

a No

b No

c No

d Yes

6

a The constant is the same.

b All of the graphs cross the y-axis at the same point.

c Equations of the form $y = mx + b$ that have the same value of b generate graphs that cross the y-axis at the same point.

7

a $y = 1$

b b

8

a

i $x = 0$

ii $y = 0$

b

i $x = 3$

ii $y = 4$

c

i There is no x-intercept.

ii $y = -1$

9

a No

b No

c Yes

d Yes

e No

f Yes

10 Positive

11

a Yes b Yes c No d No

e Yes

12

a $y = 5$ b $y = 0$ c $y = 10$ d $y = 1$

13

a Yes b No c Yes d Yes

e Yes

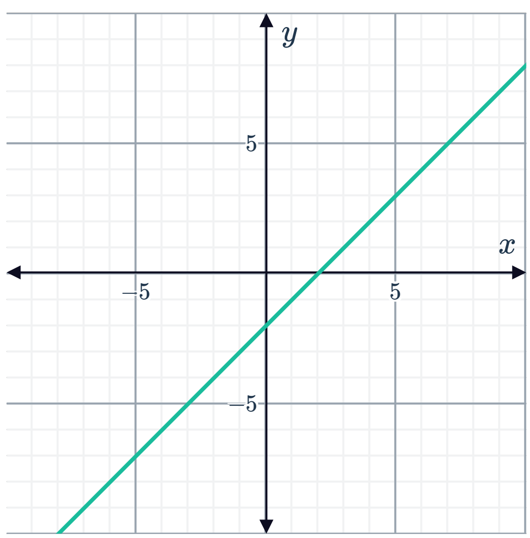
14

a $(4, 0)$ b $(0, -4)$ c $(0, -10)$

15

a $(4, 0)$ b $(0, 4)$ c $(0, 14)$

16



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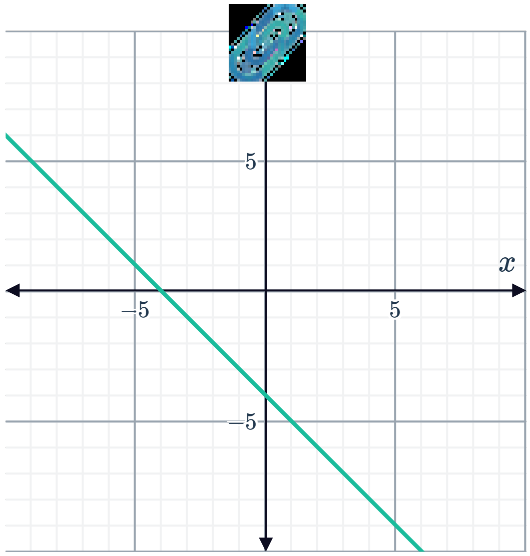

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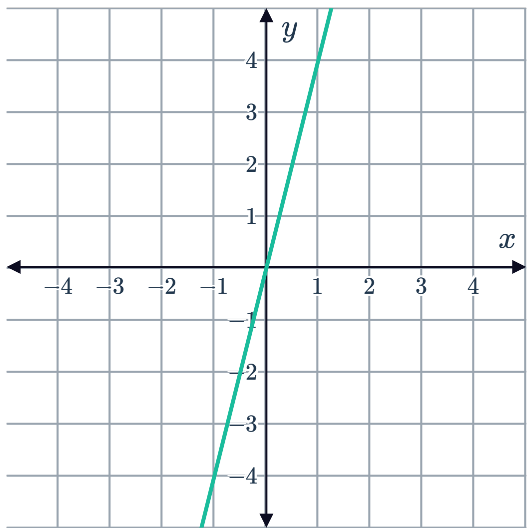


19

- **a** Yes
- **b** Yes
- **c** No
- **d** No
- **e** No

20

- **a** Yes
- **b** No
- **c** Yes
- **d** No
- **e** No
- **f** No

21 $(1, 3)$

22

- **a** Increasing
- **b** Flatter
- **c**

23

a

i Yes **ii** No **iii** No **iv** Yes

v Yes

b

i No **ii** Yes **iii** Yes

c $x = 8$

24

a Decreasing

b Steeper

c

i Yes **ii** Yes **iii** No **iv** No

v No

25

a Statement 2: The set of points lie on a horizontal line.

b Statement 2: The set of points lie on a horizontal line.

c Statement 1: The set of points lie on a vertical line.

d Statement 3: The set of points lie on an increasing line.

e Statement 1: The set of points lie on a vertical line.

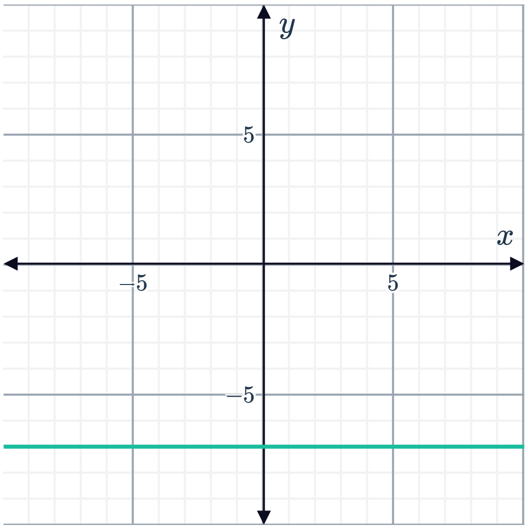
f Statement 4: The set of points lie on a decreasing line.

26

a A horizontal line **b** A vertical line

27

a



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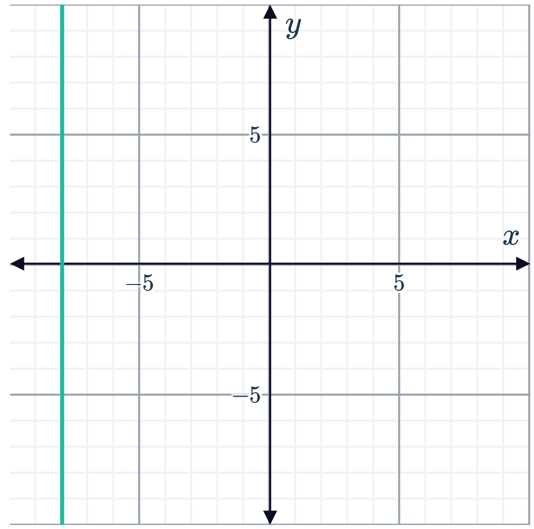


28

a $x=-8$

b None

29

a $y=-4$

b None

30

a $x=-3$

b $y=4$

31 $y=-4.5$

32 x-axis

33 y-axis

34

a $y=13,\ y=-7$

b $x=5,\ x=-9$

35 $y=0,\ y=-4$

36

a $\$2.50$

b A fixed cost such as a toll charge

Additional questions

37

a The transformation from $f(x)$ to $g(x)$ is a vertical translation.

b The transformation from $f(x)$ to $g(x)$ is making the line steeper.

c The transformation from $f(x)$ to $g(x)$ is a reflection across the x-axis.

38 When k is negative, the transformation will involve a reflection across the x-axis.

39

a Vertical translation

b Making the line steeper

c Making the line less steep

d Making the line less steep

e Reflection

f Vertical translation

40

a Vertical translation

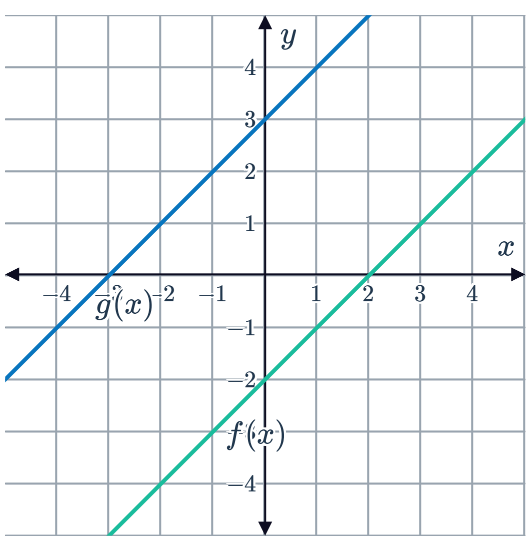
b Making the line steeper

c Making the line less steep and reflection across the x-axis

d Vertical translation

41

a



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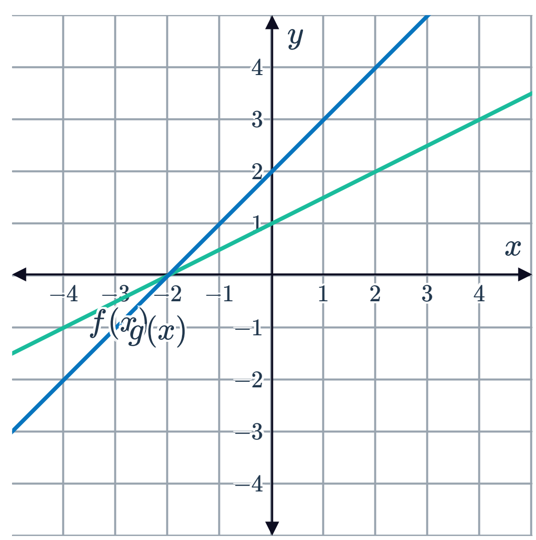

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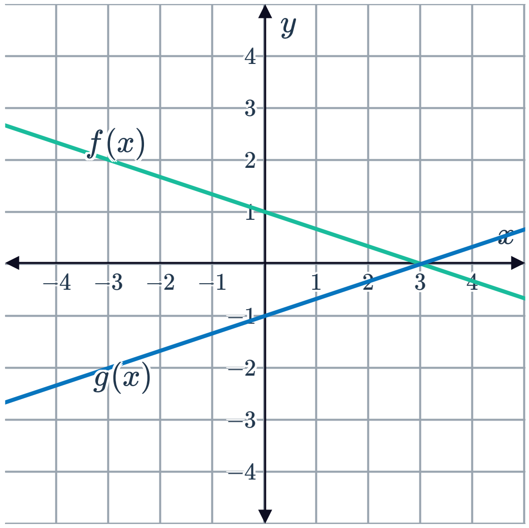

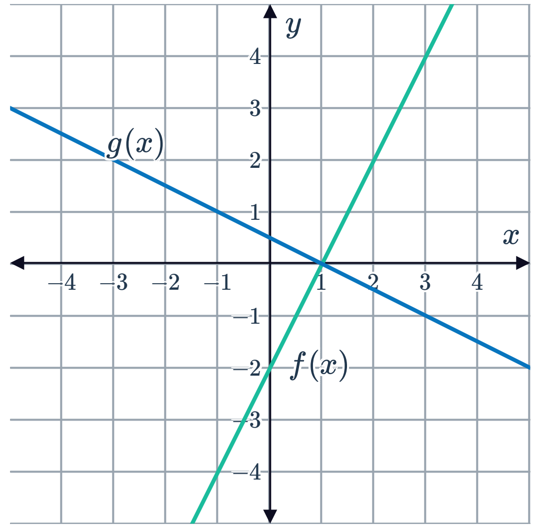

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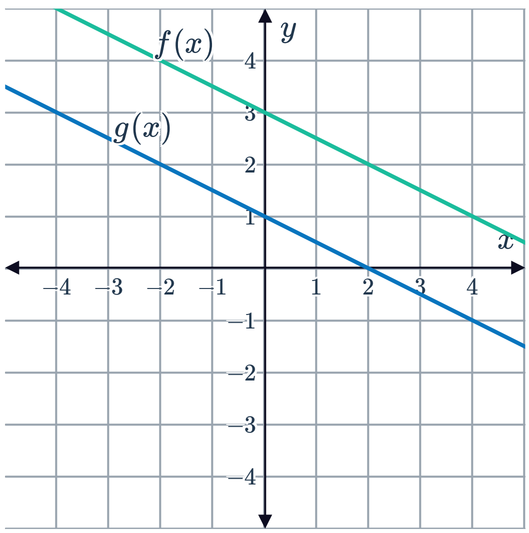

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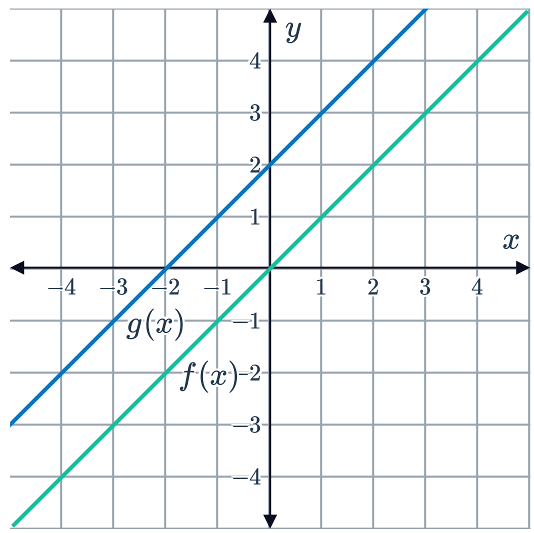

42

a The graph of $g(x)$ is the vertical translation of 8 units downwards from the graph of $f(x)$.

b The graph of $g(x)$ is the vertical translation of $\frac{1}{2}$ units upwards from the graph of $f(x)$.

c The graph of $g(x)$ is steeper than the graph of $f(x)$ by a factor of -3 and a reflection across the x-axis.

d The graph of $g(x)$ is making the graph of $f(x)$ steeper by a factor of $\frac{1}{4}$. This is equivalent to making the line less steep by a factor of 4.

43

a Oreste reflected the graph of $f(x)$ across the y-axis instead of the x-axis.

b If $f(x)$ is a linear function passing through the origin, then a reflection across the y-axis has the same effect as a reflection across the x-axis. If this is the case, Oreste's error would still give the correct answer.